

6	a(i)	6	A1
	(ii)1.	maximum photon energy = $13.6 - 0.85 = 12.75\text{eV}$ maximum kinetic energy = maximum photon energy – work function $= 12.75 - 5.6$ $= 7.15$ $= 7.2\text{ eV}$	C1 A1
	(ii)2.	photon energy = hc / λ $\lambda = (6.63 \times 10^{-34} \times 3.00 \times 10^8) / [(13.6 - 0.85) \times 1.60 \times 10^{-19}]$ $= 9.75 \times 10^{-8} = 9.8 \times 10^{-8}\text{ m}$	C1 A1

	6 b	Part Question (6b) related to CLT to be removed	
		The energy released that is converted into KE of the products is constant.	B1
		The total momentum of the system before and after the reaction is conserved.	B1
		Since the electron is found to have a range of KE, this means that there must be a third element present in order for COE and COM to be upheld	B1
		Total:	8